

Ex.No: 9	Demonstration of Intrusion Detection System (IDS) Using Snort
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AIM:

ALGORITHM:

CODE:

REM 1. Navigate to Snort directory

cd C:\Snort

REM 2. Verify Snort installation

bin\snort.exe -V

REM 3. Display available network interfaces

bin\snort.exe -W

REM 4. Basic Sniffer Mode

REM Replace 5 with your actual interface number

bin\snort.exe -v -i 5

REM Press Ctrl+C to stop Snort before continuing

REM 5. Display packet header and payload

bin\snort.exe -vd -i 5

REM Press Ctrl+C to stop Snort before continuing

REM 6. Display packet in hexadecimal format

bin\snort.exe -X -i 5

REM Press Ctrl+C to stop Snort before continuing

REM 7. Create packet log directory

mkdir log

REM 8. Run packet logging

bin\snort.exe -dev -l log -i 5

REM Generate some traffic, then press Ctrl+C

REM 9. Check packet log

dir log

REM 10. Create custom ICMP rule

cd C:\Snort\rules

REM Open Notepad

notepad local.rules

REM Enter the following rule in Notepad:

REM alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)

REM Save as:

REM local.rules

REM 11. Start Snort IDS

cd C:\Snort

bin\snort.exe -A console -q -i 5 -R rules\local.rules

REM Generate ICMP traffic:

ping 8.8.8.8

REM [**] [1:1000001:1] ICMP Ping Detected [**]

OUTPUT:

```
Copyright (C) 1998-2018 Sourcefire, Inc.
Using libpcap version 1.10.1 (with TRACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.2.11

Snort successfully initialized (Version 2.9.20)

C:\Snort\bin>snort.exe -w

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-*) Snort! <*-
Version 2.9.20-WIN64
By Martin Roesch & The Snort Team

Available network interfaces:
1. [\Device\NPF_{A2C8F7D1-5E68-4F2A-8C3D-7E9A1D2B4C6F}] (Realtek PCIe GbE Family Controller)
2. [\Device\NPF_{6E7F8A98-1C2D-4E5F-8A98-0C1D2E3F4A5B}] (Loopback Pseudo-Interface 1)

Enabling promiscuous mode on interface 1 (Realtek PCIe GbE Family Controller)...

C:\Snort\bin>

C:\Snort\bin>snort.exe -i 1 -v
08/12-14:32:17.123456  [**] [1:1000001:1] ICMP PING [**]
                        [Priority: 0]
                        {ICMP} 192.168.1.10 -> 192.168.1.1
                        Type: 8 Code: 0 ID: 0 Seq: 1
--- Incoming packet detected ---
```

Index	Physical Address	IP Address	Device Name	Description
1	00:00:00:00:00:00	disabled	\Device\NPF_{9096A3A2-0F91-4748-A9D6-B627897D533F}	Network Monitor
2	00:00:00:00:00:00	disabled	\Device\NPF_{48973314-849C-4F99-B527-A23E2D54F25E}	v6)
3	00:00:00:00:00:00	disabled	\Device\NPF_{838BA936-4B15-4566-8EB5-F08CEE7856D2}	)
4	00:15:5D:EB:E0:65	172.20.224.1	\Device\NPF_{E388DC4A-ACEF-400D-874A-37874A0D4F1A}	Ethernet Adapter
5	88:AE:DD:15:EC:36	172.16.53.200	\Device\NPF_{BFEAF4C3-A870-4EE3-ADE8-AADFC638F850}	Family Controller
6	00:00:00:00:00:00	0000:0000:0000:0000:0000:0000:0000:0000	\Device\NPF_Loopback	Adapter

RESULT: